

Organisms & Populations

Topic -Population Attributes, Population Growth, Population Interaction

1. Read the statements carefully and find how many are incorrect with respect to population interaction?

(A) Mutualism is found between Ophrys and Colpa aurea

(B) Cattle and egrets show proto cooperation

(C) Crocodile and plover bird show commensalism

(D) Interaction between Balanus and Chthamalus is competitive coexistence

- (1) 4
- (2) 3
- (3) 2
- (4) 1

2. Which of the following statement is correct?

- (1) The human liver fluke depends on one intermediate host to complete its life cycle.
- (2) Parasites that feed on the external surface of the host organism are called endoparasites.
- (3) Ectoparasites are those that live inside the host body at different sites.
- (4) Brood parasitism in birds is a fascinating example of parasitism.

3. Carrying capacity is denoted as

- (1) r
- (2) N
- (3) K
- (4) I

4. If N is the population density at time t, then its density at time t + 1 is-

- (1) $N_{t+1} = N_t + (B + E) - (D + I)$
- (2) $N_{t+1} = N_t + (B + D) - (E + I)$
- (3) $N_{t+1} = N_t - (B + I) - (D + E)$
- (4) $N_{t+1} = N_t + (B + I) - (D + E)$

5. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R) .

Assertion (A) : The interaction where one species is benefited and the other is neither benefited nor harmed, is called commensalism.

Reason (R) : An orchid growing as an epiphyte on a mango branch is an example of commensalism.

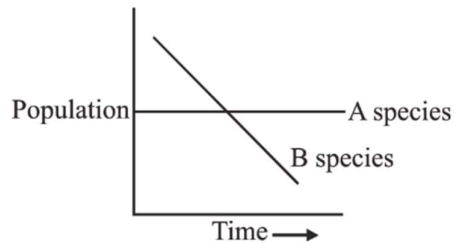
Choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) both are correct and (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is wrong.
- (3) (A) is incorrect but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

6. If in pond there are 20 lotus plants in last year & through reproduction 8 new plants are added, then the birth rate is

- (1) 0.8 offspring per lotus per year
- (2) 0.2 offspring per lotus per year
- (3) 0.4 offspring per lotus per year
- (4) 0.6 offspring per lotus per year

7. Which type of interaction is shown by below graph between species A and B ?



- (1) Mutualism
- (2) Commensalism
- (3) Competition
- (4) Amensalism

8. Gause's principle of competitive exclusion states that:

- (1) More abundant species will exclude the less abundant through competition
- (2) Larger organism will exclude smaller one
- (3) No two closely related species can occupy same niche indefinitely for the same limiting resources
- (4) Both A and B

9. Read the following statement w.r.t. population interaction

- (A) Some species of insects and frogs are cryptically coloured to avoid being detected easily by the predator
- (B) Lichens represent an mutualistic relationship between a fungus and photosynthesising algae.
- (C) Amensalism is 0, – type of interspecific interaction

- (1) Only A, C is correct
- (2) Only B, C is correct
- (3) Only C is correct
- (4) All are correct

10. Which of the following equation correctly represents species-area relationship ?

- (1) $\log C = \log S + Z \log A$
- (2) $\log A = \log C + Z \log S$
- (3) $\log A = \log S + Z \log C$
- (4) $\log S = \log C + Z \log A$

11. Mark the incorrect match w.r.t. population interactions

- (1) Amensalism : +, 0
- (2) Predation : +, –
- (3) Parasitism : +, –
- (4) Protocooperation : +, +

12. Carrying capacity is the capacity of

- (1) Habitat that has resources to sustain certain number of individuals
- (2) Population to reproduce and complete
- (3) Population to reproduce
- (4) Individuals to fit among the natural environment

13. Statement-I : Immigration is the number of individuals of same species who left the habitat and gone elsewhere during the time period under consideration.

Statement-II : Natality and emigration contribute to an increase in population density.

- (1) Both Statement-I and II are incorrect.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and II are correct.

14. Identify the correct set of statements :

(a) An orchid growing as an epiphyte on a mango tree is an example of commensalism.

(b) Lichens represent an intimate mutualistic relationship between a photosynthetic fungus and saprophytic algae.

(c) Sea anemone and Clown fish show prey-predator relationship.

(d) Gause's exclusion principle justifies the co-existence of two closely related species.

Choose the correct answer from options given below :

- (1) a only
- (2) a, b and d
- (3) a, c and d
- (4) a, b and c

15. Which of the following statement /s is not related to logistic growth ?

- (1) Environmental resistance suddenly becomes effective
- (2) Exponential phase is followed by stationary phase
- (3) Mass mortality and population crash occurs
- (4) More than one option is correct

16. The life cycle of endoparasite are more complex why?

- (1) Because of their complex morphology
- (2) Because of their food habit
- (3) Because of their extreme specialization
- (4) All of these

17. Mutualism occurs between

- (1) Butterfly and flower
- (2) E.coli and man
- (3) Zoochlorella and Hydra
- (4) Hermit crab and Sea Anemone

18. An orchid resembling female insect with ability to get pollinated is due to phenomenon of

- (1) Mimicry
- (2) Pseudocopulation
- (3) Pseudopollination
- (4) Parthenogenesis

19. Which parameter is used for founding population density of banyan tree?

- (1) Relative density
- (2) Population count in number
- (3) Biomass method
- (4) Number of prop roots

20. Given below are two statements;

Assertion :- Cuscuta is a parasite commonly found on hedge plants.

Reason :- Cuscuta has lost its chlorophyll and leaves in the course of evolution.

Choose the correct answer

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.



ANSWER KEY

1. (2) 2. (4) 3. (3) 4. (4) 5. (1) 6. (3) 7. (4) 8. (3) 9. (4) 10. (4)
11. (1) 12. (1) 13. (1) 14. (1) 15. (4) 16. (3) 17. (3) 18. (2) 19. (3) 20. (1)

